

in the schematic to correctly import it to the PCB tool.

When you import the schematic to PCB tool, you will see footprints for each component in the schematic, which are already connected through rats nest so that you don't make any mistake while routing.

4. Routing and export. You can now start routing by selecting various routing options from the tool. Footprint libraries of both the tools are also good enough and new footprints can be made very quickly. Routing and other features are very easy to use and learn. Netlist, Gerber and BOM can be directly generated for production.

Q2. I am an electronics and telecommunication engineering student, and EFY reader for the last two years. Please suggest me some good books for VLSI design.

Ashish Kumbhare

Through e-mail

A2. There are hundreds of books available on VLSI design but selection of the right book for you depends on the course that you want to cover and prerequisite skills for a particular book. Some books might expect you to already have some basic knowledge of the topic.

Mentioned below are some good books on VLSI design:

1. 'Digital Integrated Circuits' by Jan M. Rabaey, Anantha Chandrakasan, Borivoje Nikolic
2. 'CMOS VLSI Design: A Circuits and Systems Perspective' by Neil H. E. Weste and David Money Harris
3. 'Principles of CMOS VLSI Design' by Neil H. E. Weste and Kamran Eshraghian
4. 'Basic VLSI Design: Systems and Circuits' by Douglas, A. Pucknell and Kamran Eshraghian
5. 'Modern VLSI Design: IP-based Design' by Wayne Wolf
6. 'Introduction to VLSI Circuits and Systems' by John P. Uyemura

Go through the prerequisites and index of each book to decide one for you. However, do not restrict yourself to one book only as it might not cover your complete syllabus.

Q3. What's the difference between cumulative and differential compound motors?

Vamshi Kittu

Through e-mail

A3. Compound motors are designed as a combination of series DC motors and shunt DC motors to attain the characteristics of both types of motors (refer Fig. 2). This combination can achieve the torque characteristics of series DC motors and constant-speed characteristics of shunt DC motors.

A compound motor has both series winding and shunt winding. One winding is connected in series with the armature as in a series DC motor. Another winding is connected in shunt with the armature as in a shunt DC motor.

Depending on the relative polarity of series and shunt windings, there are majorly three classifications of DC compound motors: Cumulative compound motors, differential compound motors and compound interpole motors.

Cumulative compound motors. The polarity of the shunt winding is such that it adds to the series fields in cumulative compound motors. This shunt winding can be either short shunt or long shunt. In short shunt the shunt is parallel with only the armature field, whereas in long shunt the shunt is parallel to both the armature and the series field. Since shunt windings are done in such a way as to assist both the armature and series field producing a cumulative effect, the motor is called cumulative compound motor.

Cumulative motors give a high starting torque like a series motor, and reasonable good speed regulation at high speeds like a shunt DC motor. These can start with huge loads and run smoothly. As cumulative motors offer the best of both series and shunt motors, these are widely used for different applications.

Differential compound motors. In differential compound motors, polarities of the armature and the shunt field oppose each other. In this type of motor, negative terminal of the shunt field is connected to the positive terminal of the armature.

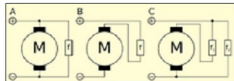


Fig. 2: Shunt, series and compound DC motors

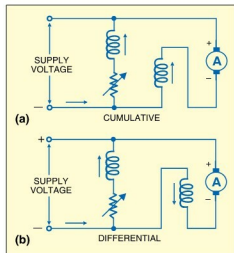


Fig. 3: Cumulative and differential DC motors

In differential compound motors, magnetic fields of the shunt winding oppose the armature magnetic fields and the series fields. This kind of differential winding provides different torque and speed characteristics. Here as the shunt field is producing an opposite effect, it is unlike a shunt motor. So when the load is reduced, differential compound motor behaves more like a series motor and tends to overspeed. When the load is increased, its speed reduces drastically.

Compound interpole motors. Compound interpole motors have additional interpoles in series with the armature. Interpoles are connected in series in between the series winding and the armature. Physically, these are placed beside the series coils in the stator. The polarities of interpoles and series fields are the same and they assist each other. In interpole motors brushes last longer, so it is not necessary to cut down the armature often. Overall, interpoles help to improve smooth functioning of the motor, prolonging its life.

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